The Many Telecom Applications of Asterisk

Over the course of the past few articles in the series, we got familiar with the general Asterisk environment, the typical hardware involved and the implementation of a basic IP PBX, which is the most important Asterisk application. In this article, the author outlines multiple applications that can be based on the Asterisk platform, which will help to understand its potential as a telecommunications platform.

Asterisk has been conceptualised as a generic switching platform, on which a variety of telecom applications can be run. Let's list and describe these.

IP PBX (IP Private Branch eXchange)

IP PBX is IP based equipment, which can be used for internal communication within companies (commonly known as an intercom) and for employees to communicate with the outside world. The telephones in the company are connected to the PBX to enable internal communication. Trunk lines, which could be analogue, digital, GSM or IP, will be connected to the PBX to enable external communication. Over the previous articles, I have mainly focused on Asterisk as a PBX, and therefore further elaborate details have been avoided here.

IVR (Interactive Voice Response)

Today, it is common to obtain a menu when we call our bank or insurance company. A recorded voice will ask us about the intended transaction, personal details, etc, and provide us appropriate information or direct us towards a human agent.

A common IVR is the welcome menu at most companies. When a customer calls the central number, the IVR is activated. As first, it plays a message like, "Welcome to *astTECS. Please press 1 if you want to talk to a sales person, press 2 for technical assistance or 3 for any other matter." The number the customer presses is transmitted as a dual tone multi-frequency (DTMF) signal from the phone to the IVR server. Depending on the input, the call is transferred to the 'Sales' queue, the 'Technical' queue or the 'Admin' queue.

Many companies have reduced their manpower requirements by deploying powerful IVRs. The call centre IVR for banks is a typical example. Most of them play the welcome message, ask users to select the language, carry out the authentication in the selected language and connect to the respective department. In many cases, where the customer requires just the account information or card status, the process is completed without any manual intervention. The IVR script itself will query the database